Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
11.	(a)	(i)		**Indicative content** • add acid (not sulfuric/hydrochloric) to prove if carbonate present • if fizzing/effervescence carbonate present • if carbonate present, add acid until no more fizzing • divide solution into two (accept divide solution before adding acid) • add barium chloride to one half • precipitate shows that sulfate is present • add siver nitrate to other half • precipitate shows that chloride is present	2	2	2	6		6
				5-6 marks Devises a plan with relevant observations and conclusions that unambiguously shows that one, two or three anions are present. *The candidate constructs a relevant, coherent and logically structured method including all key elements of the indicative content. A sustained and substantiated line of reasoning is evident and scientific conventions and vocabulary are used accurately throughout.* **3-4 marks** Devises a plan with relevant observations and conclusions that shows that two or three anions could be present. *The candidate constructs a coherent account including most of the key elements of the indicative content. Some reasoning is evident in the linking of key points and use of scientific conventions and vocabulary are generally sound.* **1-2 marks** Devises a simple plan with some observations and conclusions that shows that one of the anions could be present. *The candidate attempts to link at least two relevant points from the indicative content. Coherence is limited by omission and/or inclusion of irrelevant material. There is some evidence of appropriate use of scientific conventions and vocabulary.* **0 marks** *The candidate does not make any attempt or give an answer worthy of credit.*						

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Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
					Marks available					
		(ii)		incorrect since other cations form soluble salts with these anions (1) carry out flame test – if flame turns yellow cation is sodium (1)	 1		1	 2		 1
	(b)	(i)		**A** is simple molecular (1) **D** is ionic (1) **B** is **probably** giant molecular but **could be** ionic since not all ionic substances are soluble (1) **C** is **probably** metallic but **could be** giant molecular (graphite) since graphite conducts electricity (1)	2		2	4		
		(ii)		**B** - use higher temperatures to melt the solid and test the molten substance for electrical conductance (1) if **B** conducts then ionic / if **B** does not conduct then giant molecular (1) **OR** **C** - add acid (and warm) and see if effervescence forms (1) if **C** effervesces then metal / if **C** does not effervesce then graphite (1) credit sensible alternative physical tests e.g. hardness of **B**, malleability of **C**		1	1	2		2
		(iii)		can be difficult to get powdered metal to conduct electricity / powdered metal not shiny / not sonorous / not malleable		1		1		1
				Question 11 total	**5**	**4**	**6**	**15**	**0**	**10**

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